

Advanced Database Administration I

Planning and Creating an Oracle9i Database

Goals:

The goal of this assignment is to demonstrate the importance of planning when building a real-world database using Oracle 9i Release2 as a database management system. It helps the students to acquire the necessary knowledge at the field of database administration and apply this knowledge on a practical case.



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Introduction:

The Oracle Database (commonly referred to as *Oracle RDBMS* or simply as *Oracle*) is a relational database management system (RDBMS) produced and marketed by Oracle Corporation. As of 2010, Oracle has been a major presence in database computing for many years. Larry Ellison and his friends and former co-workers Bob Miner and Ed Oates started the consultancy Software Development Laboratories (SDL) in 1977. SDL developed the original version of the Oracle software. The name *Oracle* comes from the code-name of a CIA-funded project Ellison had worked on while previously employed by Ampex.

Homework text:

You've been asked to create an Oracle database that meets the following requirements with best performance:

1. Your server currently has four disk drives (C:, D:, E:, F:)
2. Oracle R2 is installed, and the installation is OFA-compliant
3. Your database name and system identifier are as the same as your group number (ex: G10)
4. SGA_MAX_SIZE = 256 MB
5. DB_CACHE_SIZE = 60 MB
6. SHARED_POOL_SIZE = 48 MB
7. No Java option is required with the database
8. Shared Server mode is used
9. DB_BLOCK_SIZE = 8 K
10. The database should not accept more than 50 datafiles, 3 log groups, and 3 log members during its lifetime
11. Automatic undo management
12. Tablespaces are

TABLESPACE	DATAFILES	TYPE
system	2	Dictionary managed
temp_tbls	1	Locally managed
undo_tbls	1	Locally managed
data_tbls	4	Locally managed
index_tbls	1	Locally managed

13. DEFAULT profile is applied to all users
14. The database should be available 7/7 days, and 24/24 hours
15. No data loss is allowed during the database lifetime

Requests and Answers:

Please answer the following questions with respect to all the previous requirements:

Question 1:

What are the values for the environment variables :

	variables	value
1.1	ORACLE_BASE	C:\oracle
1.2	ORACLE_HOME	C:\oracle\ora92
1.3	ORACLE_SID	G8

Question 2:

What is the best value for LARGE_POOL_SIZE, justify your answer

LARGE_POOL_SIZE Specifies the size of the large pool allocation heap, which is used by Shared Server for session memory, parallel execution for message buffers, and RMAN backup and recovery for disk I/O buffers. Range of Values: 600K (minimum); >= 20000M (maximum is operating system specific). Default Value: 0, unless parallel execution or DBWR_IO_SLAVES are configured.

A granule is a unit of contiguous virtual memory allocation. The size of a granule depends on the estimated total SGA size whose calculation is based on the value of the parameter SGA_MAX_SIZE.

- Granule =16 MB while SGA size=256 is larger than 128 MB.
- The components (buffer cache, shared pool, and large pool) are allowed to grow and shrink based on granule boundaries. For each component which owns granules, the number of granules allocated to the component. At instance startup, the Oracle server allocates granule entries, one for each granule to support SGA_MAX_SIZE bytes of address space. As startup continues, each component acquires as many granules as it requires. The minimum SGA configuration is three granules (one granule for fixed SGA) includes redo buffers; one granule for buffer cache; one granule for shared pool).

LARGE_POOL_SIZE=16 M

Question 3:

What are the Minimum required initialization parameters that should be used to create the database? List these parameters with suitable values

No	Parameter value	Parameter name
1	DB_NAME	G8
2	INSTANCE_NAME	G8
3	CONTROL_FILES	C:\oracle\oradata\G8\control01.ctl
		D:\oracle\oradata\G8\control02.ctl
		E:\oracle\oradata\G8\control03.ctl
4	BACKGROUND_DUMP_DEST	C:\oracle\admin\G8\bdump
5	USER_DUMP_DEST	C:\oracle\admin\G8\udump
6	DB_FILES	50
7	DB_BLOCK_SIZE	8192 Byte
8	DB_CACHE_SIZE	62,914,560
9	Log_buffer	524288
10	SHARED_POOL_SIZE	50331648
11	COMPATIBLE	9.2.0.0.0
12	large_pool_size	16 M
13	UNDO_MANAGEMENT	Auto
14	log_archive_start	TRUE

Question 4:

Write the CREATE DATABASE script

```
CREATE DATABASE G8
  MAXLOGFILES 3
  MAXLOGMEMBERS 3
  MAXDATAFILES 50
  DATAFILE
    'C:\oracle\oradata\G8\system01.dbf' SIZE 250M REUSE
  AUTOEXTEND ON NEXT 10240K MAXSIZE UNLIMITED,
  'C:\oracle\oradata\G8\system11.dbf' SIZE 5M
  DEFAULT TEMPORARY TABLESPACE TEMP_TBLS tempfile
  'D:\oracle\oradata\G8\TEMP_TBLS.dbf' SIZE 5M
  UNDO TABLESPACE "UNDO_TBLS" DATAFILE
  'E:\oracle\oradata\G8\UNDO_TBLS.dbf' SIZE 5M
  CHARACTER SET AL32UTF8 NATIONAL CHARACTER SET AL16UTF16
  LOGFILE GROUP 1 (
    'D:\oracle\oradata\G8\redo01.log',
    'D:\oracle\oradata\G8\redo11.log',
    'D:\oracle\oradata\G8\redo21.log') SIZE 102400K,
  GROUP 2 (
    'E:\oracle\oradata\G8\redo02.log',
    'E:\oracle\oradata\G8\redo12.log',
    'E:\oracle\oradata\G8\redo22.log') SIZE 102400K,
  GROUP 3 (
    'F:\oracle\oradata\G8\redo03.log',
    'F:\oracle\oradata\G8\redo13.log',
    'F:\oracle\oradata\G8\redo23.log') SIZE 102400K;
```

Creating Tablespaces and Datafiles:

```
CREATE TABLESPACE "DATA_TBLS" LOGGING DATAFILE
  'F:\oracle\oradata\G8\DATA_TBLS01.dbf' SIZE 5M,
  'F:\oracle\oradata\G8\DATA_TBLS11.dbf' SIZE 5M,
  'F:\oracle\oradata\G8\DATA_TBLS21.dbf' SIZE 5M,
  'F:\oracle\oradata\G8\DATA_TBLS31.dbf' SIZE 5M EXTENT
  MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO ;

CREATE TABLESPACE "INDEX_TBLS" LOGGING DATAFILE
  'F:\oracle\oradata\G8\INDEX_TBLS.dbf' SIZE 5M EXTENT
  MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO ;
```

Question 5:

Write all the instructions necessary to achieve the following;

- 1.1 Each user is allowed to have 3 concurrent sessions:

```
ALTER PROFILE DEFAULT LIMIT
SESSIONS_PER_USER 3;
```

Then when we create a new User account we won't apply any profile to this user, so the user can apply default profile setting.

To show value of session_per_user parameter:

```
SELECT RESOURCE_NAME, LIMIT FROM DBA_PROFILES WHERE
PROFILE='DEFAULT';
```

- 1.2 Not more than 10 concurrent users are allowed to access the database

Use the STARTUP command to restrict access to a database:

A restricted session is useful, for example, when you perform structure maintenance or a database export and import. The database can be started in restricted mode, so that it is available only to users with the RESTRICTED SESSION privilege. The database can also be put in restricted mode by using the ALTER SYSTEM SQL command:

```
ALTER SYSTEM ENABLE RESTRICTED SESSION
```

and we give restrict privilege for 10 users

Question 6:

What is your recommended database structure for the best availability, and with respect to all the previous conditions, justify your answer. N.B: You should consider names of files (datafiles, redolog files, and control files) and their locations

No	Physical File Name	Physical File Location
1	SYSTEM01.DBF	C:\oracle\oradata\G8\system01.dbf
2	SYSTEM02.DBF	C:\oracle\oradata\G8\system11.dbf
3	DATA_TBLS01.DBF	F:\oracle\oradata\G8\DATA_TBLS01.dbf
4	DATA_TBLS11.DBF	F:\oracle\oradata\G8\DATA_TBLS11.dbf
5	DATA_TBLS21.DBF	F:\oracle\oradata\G8\DATA_TBLS21.dbf
6	DATA_TBLS31.DBF	F:\oracle\oradata\G8\DATA_TBLS31.dbf
7	INDEX_TBLS01.DBF	F:\oracle\oradata\G8\INDEX_TBLS.dbf
8	TEMP_TBLS01.DBF	D:\oracle\oradata\G8\TEMP_TBLS.dbf
9	UNDO_TBLS01.DBF	E:\oracle\oradata\G8\UNDO_TBLS.dbf
10	CONTROL01.CTL	C:\oracle\oradata\G8\CONTROL01.CTL
11	CONTROL02.CTL	C:\oracle\oradata\G8\CONTROL02.CTL
12	CONTROL03.CTL	C:\oracle\oradata\G8\CONTROL03.CTL
13	REDO01.LOG	D:\oracle\oradata\G8\redo01.log
14	REDO11.LOG	D:\oracle\oradata\G8\redo11.log
15	REDO21.LOG	D:\oracle\oradata\G8\redo21.log
16	REDO02.LOG	E:\oracle\oradata\G8\redo02.log
17	REDO12.LOG	E:\oracle\oradata\G8\redo12.log
18	REDO22.LOG	E:\oracle\oradata\G8\redo22.log
19	REDO03.LOG	F:\oracle\oradata\G8\redo03.log
20	REDO13.LOG	F:\oracle\oradata\G8\redo13.log
21	REDO23.LOG	F:\oracle\oradata\G8\redo23.log

Question 7:

How can you satisfy conditions 14 & 15, justify your answer

The database should be available 7/7 days, and 24/24 hours:

Database can't startup normally without control files, so we used multiplexing on Control file and create three copies of these files on several Disk Drive (C,D,E). When C Drive is damaged we can use the other control files which are located in D or E drives. We should use hot backup which doesn't need to shutdown the database.

No data loss is allowed during the database lifetime

Best way to save data from losing is to apply Archiving mode like what we did.

`log_archive_dest_1='LOCATION=F:\oracle\oradata\G8\archive'`

`log_archive_format=%t_%s.dbf`

`log_archive_start=true`

We should save a copy of our database by doing backup.

Question 8:

Suppose that OMF is used by the administrator;

1.1. What are the required initialization parameters:

```
ALTER SYSTEM SET
db_create_file_dest = 'C:\oracle\oradata\G8';
ALTER SYSTEM SET
DB_CREATE_ONLINE_LOG_DEST_1= 'D:\oracle\oradata\G8';
DB_CREATE_ONLINE_LOG_DEST_2= 'E:\oracle\oradata\G8';
DB_CREATE_ONLINE_LOG_DEST_3= 'F:\oracle\oradata\G8';
```


1.2. What would be the CREATE DATABASE script?

```
CREATE DATABASE G8
MAXLOGFILES 3
MAXLOGMEMBERS 3
MAXDATAFILES 50

DATAFILE
'C:\oracle\oradata\G8\system01.dbf' SIZE 250M REUSE
AUTOEXTEND ON NEXT 10240K MAXSIZE UNLIMITED,
'C:\oracle\oradata\G8\system11.dbf' SIZE 5M

DEFAULT TEMPORARY TABLESPACE TEMP_TBLS TEMPFILE
UNDO TABLESPACE "UNDO_TBLS" DATAFILE
CHARACTER SET AL32UTF8
NATIONAL CHARACTER SET AL16UTF16
LOGFILE GROUP 1 ,GROUP 2,GROUP 3
Now Creating Data Files
CREATE TABLESPACE "DATA_TBLS"
CREATE TABLESPACE "INDEX_TBLS"
```

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